ETSI EN 300 328 V2.2.2 (2019-07)

TEST REPORT

For

Quanzhou STAYTRUE Electronic Technology Co., Ltd

Unit 1A-2, 1st Floor, Building 1, Fengshu Logistics Park, No. 920, Binhu South Road, Taiwan Investment Zone, Quanzhou, Fujian Province, China

Tested Model: SDC-LE-02HD

Report Type:	**Product Name:**
Original Report	Module SDC-LE-02HD

Report Number:	2607T13127E-RF-01
Report Date:	2026-06-30
Reviewed By:	Ash Lin
Approved By:	Miles Chen
Prepared By:	Bay Area Compliance Laboratories Corp.(Fujian) 4/F., No.65 Siming Industrial Zone, Meixi Road, Tong'an District, Xiamen, Fujian, China Tel: +86-592-3200111 www.baclcorp.com.cn

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REPORT REVISION HISTORY

Number of Revisions	Report No.	Version	Issue Date	Description
0	2607T13127E-RF-01	R1V1	2026-06-30	Initial Release

GENERAL INFORMATION

Product Description for Equipment under Test (EUT)

Applicant:	Quanzhou STAYTRUE Electronic Technology Co., Ltd
Applicant Address:	Unit 1A-2, 1st Floor, Building 1, Fengshu Logistics Park, No. 920, Binhu South Road, Taiwan Investment Zone, Quanzhou, Fujian Province, China
Manufacturer:	Quanzhou STAYTRUE Electronic Technology Co., Ltd
Manufacturer Address:	Unit 1A-2, 1st Floor, Building 1, Fengshu Logistics Park, No. 920, Binhu South Road, Taiwan Investment Zone, Quanzhou, Fujian Province, China
Product Name:	Module SDC-LE-02HD
Tested Model:	SDC-LE-02HD
Series Model:	N/A
Trade Mark:	SDC-LE-02HD
Power Supply:	DC 3.3V
Maximum EIRP:	4.256dBm
RF Function:	BLE
Operating Band/Frequency:	2402-2480 MHz
Channel Number:	40
Channel Separation:	2 MHz
Modulation Type:	GFSK
★Maximum Antenna Gain:	1.5dBi
EUT Received Status:	Good
Note: *1. The maximum antenna gain is provided by the applicant.* *2. All measurement and test data in this report was gathered from production sample serial number: 3LJ5-1(RF conducted), 3LJ5-2(Radiated Spurious Emission)(Assigned by BACL (Fujian). The EUT was received on 2026-05-08).*	

Objective

This report is prepared for *Quanzhou STAYTRUE Electronic Technology Co., Ltd* in accordance with ETSI EN 300 328 V2.2.2 (2019-07), Wideband transmission systems; Data transmission equipment operating in the 2,4 GHz band; Harmonised Standard for access to radio spectrum

The objective is to determine the compliance of EUT with ETSI EN 300 328 V2.2.2 (2019-07).

Test Methodology

All measurements contained in this report were conducted with ETSI EN 300 328 V2.2.2 (2019-07).

Measurement Uncertainty

Where relevant, the following measurement uncertainty levels have been estimated for tests performed on theproduct as specified in CISPR 16-4-2. This uncertainty represents expanded uncertainty expressed at 95%confidence level using a coverage factor of k=2.
If U_{lab} is less than or equal to Ucispr, then compliance is deemed to occur if no measured disturbance level exceeds the disturbance limit.

$$u_c(y) = \sqrt{\sum_i c_i^2 \; u^2(x_i)}$$

Item	Frequency Range	U_{cispr}	U_{lab}= 2 U_c (y) （Confidence of 95%）
Radiated Spurious Emission	30MHz~1GHz	5.06dB	4.68dB
	1GHz~18GHz	5.12dB	5.00dB
Transmitter Conducted Power			±0.43dB
Power Spectral Density			2.52dB
Occupy Bandwidth			1.69%
Receiver Blocking characteristics			3.2dB
Voltage (DC)			±0.4%
Temperature			±1℃
Humidity			±5%

Test Facility

The Test site used by Bay Area Compliance Laboratories Corp. (Fujian) to collect test data is located on the Unit 302, No. 902, Meifeng South Road, Tong'an District, Xiamen, Fujian, China.

SYSTEM TEST CONFIGURATION

Test Mode and Voltage

The system was configured for testing in a typical mode (as normally used by a typical user).	
Test mode:	Test Mode 1: Transmitting Test Mode 2: Receiving
Test voltage:	DC 3.3V
Remark:	During all emission tests, the EUT was configured to measure its highest possible emission level and the worst case's test data was presented in this test report.

Description of Test Configuration

Channel list:

Channel	Frequency (MHz)	Channel	Frequency (MHz)
0	2402	20	2442
1	2404	…	…
…	…	…	…
…	…	…	…
…	…	38	2478
19	2440	39	2480

EUT was tested with channel 0, 19 and 39.

The extreme temperature test conditions which were declared by the manufacturer and the normal conditions are as below:
NT: Normal Temperature +25℃
LT: Low Temperature -40℃
HT: High Temperature +85℃

★EUT Exercise Software

RF test tool: BT_Tool.exe

Test Modes	Power Level Setting		
	Lowest Channel	Middle Channel	Highest Channel
BLE 1Mbps	6	6	6

Note: The power level setting was declared by the applicant.

Equipment Modifications

No modification was made to the EUT tested.

Support Equipment List and Details

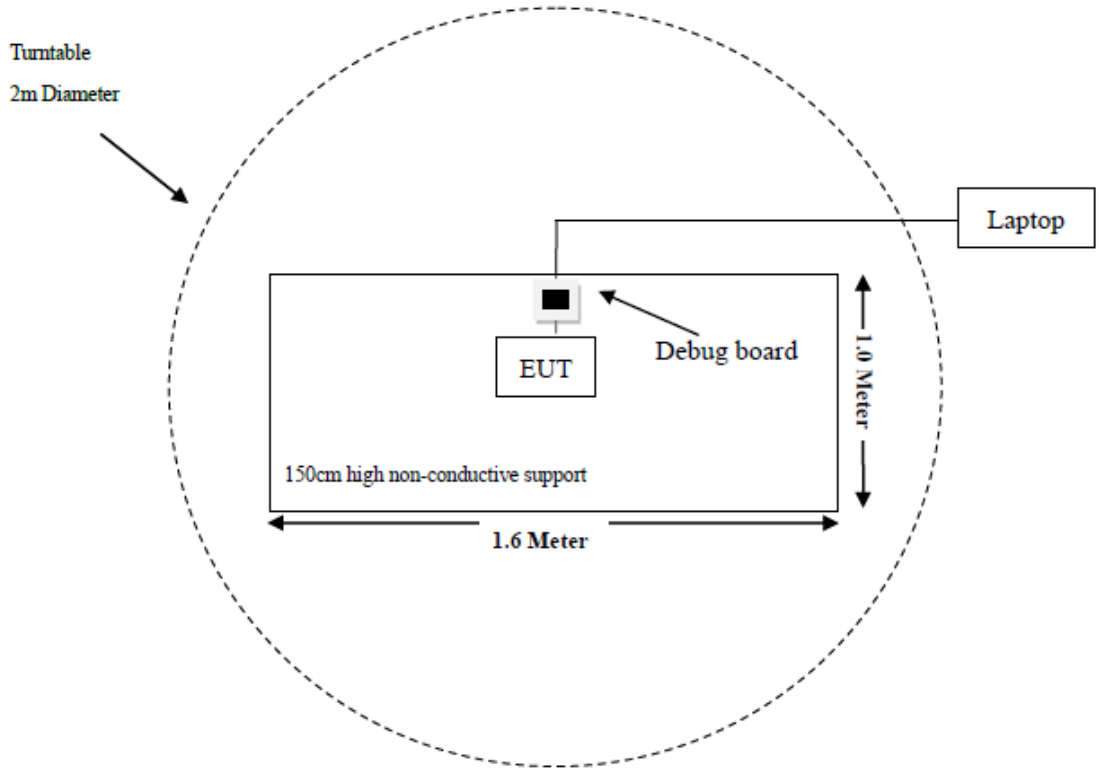
Manufacturer	Description	Model	Serial Number
Unknown	Debug board	Unknown	Unknown
Lenovo	Laptop	T480	PF1P5K4F

External Cable

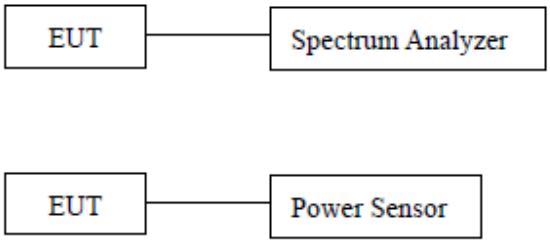
Cable Description	Length(m)	From Port	To Port
Jumper Wire	0.2	EUT	Debug board
USB cable	10	Laptop	Debug board

Block Diagram of Test Setup

Radiated Spurious Emission (30MHz-12.75GHz):



RF Conducted:



Note: The cable assembly insertion loss of 0.5dB was entered as an offset in the spectrum analyzer/power sensor. (Actual cable loss was unavailable at the time of testing, therefore loss of 0.5dB was assumed as worst case.) This was later verified to be true by laboratory.

SUMMARY OF TEST RESULTS

ETSI EN 300 328 V2.2.2 (2019-07)	Description of Test	Test Result
Clause 4.3.2.2	RF output power	Compliant
Clause 4.3.2.3	Power Spectral Density	Compliant
Clause 4.3.2.4	Duty Cycle, Tx-sequence, Tx-gap	Not Applicable (See Note 1)
Clause 4.3.2.5	Medium Utilization (MU) factor	Not Applicable (See Note 1)
Clause 4.3.2.6	Adaptivity	Not Applicable (See Note 2)
Clause 4.3.2.7	Occupied Channel Bandwidth	Compliant
Clause 4.3.2.8	Transmitter unwanted emissions in the out-of-band domain	Compliant
Clause 4.3.2.9	Transmitter unwanted emissions in the spurious domain	Compliant
Clause 4.3.2.10	Receiver spurious emissions	Compliant
Clause 4.3.2.11	Receiver Blocking	Compliant
Clause 4.3.2.12	Geo-location capability	Not Applicable (See Note 3)

Note:
1. The EUT is adaptive equipment, while this item is only for non-adaptive mode.
2. The requirement does not apply for equipment with a maximum declared RF Output power of less than 10 dBm e.i.r.p.
3. The supplier declared that the equipment is unable to perform this function.

TEST EQUIPMENT LIST

Test Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due Date
Radiated Spurious Emissions Below 1 GHz					
EMI Test Receiver	Rohde & Schwarz	ESR	103103	2026/01/21	2027/01/20
Antenna	Sunol Sciences	JB6	A122022-5	2023/07/27	2026/07/26
Dipole antenna	EMCO	3121C	9209-860	N/A	N/A
Microwave Analog Signal Generator	Agilent	N5183A	MY47420335	2026/01/21	2027/01/20
Amplifier	Sonoma	310B	120903	2026/01/22	2027/01/21
Coaxial Cable	XINHANGWEIBO	XH400T-N-4M	CC002	2026/01/20	2027/01/19
Coaxial Cable	XINHANGWEIBO	XH460B-N-2M	CC006	2026/01/20	2027/01/19
Coaxial Cable	XINHANGWEIBO	XH460B-N-12M	CC007	2026/01/20	2027/01/19
Radiated Spurious Emissions Above 1 GHz					
Spectrum Analyzer	Rohde & Schwarz	FSV40-N	102051	2026/01/21	2027/01/20
Filter Switch Unit	Decentest	DT7220FSU	DS79904	2026/01/20	2027/01/19
Multiplex Switch Test Control Set	Decentest	DT7220SCU	DS79901	2026/01/21	2027/01/20
Horn Antenna	EMCO	3115	9002-3355	2024/11/19	2027/11/18
Double Ridge Guide Horn Antenna	A.H.Systems	SAS-571	1980	2023/07/28	2026/07/27
Preamplifier	GLOBAL	1313-A100M18G	4121301	2026/01/14	2027/01/13
Coaxial Cable	XINHANGWEIBO	XH800A-N-6M	CC003	2026/01/20	2027/01/19
Coaxial Cable	XINHANGWEIBO	XH800A-N-1M	CC005	2026/01/20	2027/01/19
Microwave Analog Signal Generator	Agilent	N5183A	MY47420335	2026/01/21	2027/01/20

Test Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due Date
RF Conducted Test					
Spectrum Analyzer	Rohde & Schwarz	FSU	100405	2026/01/21	2027/01/20
constant temperature and humidity testing machine	BACL	BTH-150-60	30234	2025/07/28	2026/07/27
Microwave Analog Signal Generator	Agilent	N5183A	MY47420335	2026/01/21	2027/01/20
Wideband Radio Communication Tester	Rohde & Schwarz	CMW500	122920	2026/01/22	2027/01/21
USB Wideband Power Sensor	Boonton	55318	8934	2025/09/08	2026/09/07
2 Way power divider	narda	4426LB-2	1661	Each time	N/A
Attenuator	Agilent	8496B	MY42140574	Each time	N/A
Attenuator	HP	8494B	2550A12096	Each time	N/A
Directional coupler	narda	4226-20	08412	2026/01/22	2027/01/21
Coaxial Cable	YINSAIGE	SS405-0.5M	CC018	2025/09/25	2026/09/24
Coaxial Cable	YINSAIGE	SS405-0.5M	CC019	2025/09/25	2026/09/24

Statement of Traceability: Bay Area Compliance Laboratories Corp. (Fujian) attests that all calibrations have been performed in accordance to requirements that traceable to National Primary Standards and International System of Units (SI).

ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.2.2 – RF OUTPUT POWER

Applicable Standard

The RF output power is defined as the mean equivalent isotropically radiated power (e.i.r.p.) of the equipment during a transmission burst.

Limit

For adaptive equipment using wide band modulations other than FHSS, the maximum RF output power shall be 20 dBm.
The maximum RF output power for non-adaptive equipment shall be declared by the manufacturer and shall not exceed 20 dBm. See clause 5.4.1 m). For non-adaptive equipment using wide band modulations other than FHSS, the maximum RF output power shall be equal to or less than the value declared by the manufacturer.
This limit shall apply for any combination of power level and intended antenna assembly.

Test Procedure

The test procedure shall be as follows:

Step 1:

• Use a fast power sensor with a minimum sensitivity of -40 dBm and capable of minimum 1 MS/s.
• Use the following settings:
- Sample speed 1 MS/s or faster.
- The samples shall represent the RMS power of the signal.
- Measurement duration: For non-adaptive equipment: equal to the observation period defined in clause 4.3.1.3.2 or clause 4.3.2.4.2. For adaptive equipment, the measurement duration shall be long enough to ensure a minimum number of bursts (at least 10) is captured.

NOTE 1: For adaptive equipment, to increase the measurement accuracy, a higher number of bursts may be used.

Step 2:

• For conducted measurements on devices with one transmit chain:
- Connect the power sensor to the transmit port, sample the transmit signal and store the raw data. Use these stored samples in all following steps.
• For conducted measurements on devices with multiple transmit chains:
- Connect one power sensor to each transmit port for a synchronous measurement on all transmit ports.
- Trigger the power sensors so that they start sampling at the same time. Make sure the time difference between the samples of all sensors is less than 500 ns.
- For each individual sampling point (time domain), sum the coincident power samples of all ports and store them. Use these summed samples as the new stored data set.

Step 3:

• Find the start and stop times of each burst in the stored measurement samples.
The start and stop times are defined as the points where the power is at least 30 dB below the highest value of the stored samples in step 2.

NOTE 2: In case of insufficient sensitivity of the power sensor (e.g. in case of radiated measurements), the value of 30 dB may need to be reduced appropriately.

Step 4:

• Between the start and stop times of each individual burst calculate the RMS power over the burst using the formula below. The start and stop points shall be included.Save these P_{burst} values, as well as the start and stop times for each burst.

$$P_{burst} = \frac{1}{k}\sum_{n=1}^{k} P_{sample}(n)$$

With 'k' being the total number of samples and 'n' the actual sample number

Step 5:

• The highest of all P_{burst} values (value "A" in dBm) will be used for maximum e.i.r.p. calculations.

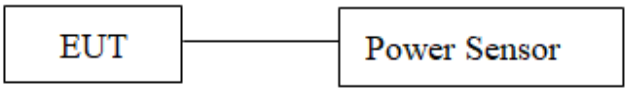
Step 6:

• Add the (stated) antenna assembly gain "G" in dBi of the individual antenna.
• In case of smart antenna systems operating in mode with beamforming (see clause 5.3.2.2.4), add the additional beamforming gain Y in dB.
• If more than one antenna assembly is intended for this power setting, the maximum overall antenna gain (G or G + Y) shall be used.
• The RF Output Power (P_{out}) shall be calculated using the formula below:

$$\mathbf{P_{out} = A + G + Y}$$

• This value, which shall comply with the limit given in clauses 4.3.1.2.3 or clause 4.3.2.2.3, shall be recorded in the test report.

Test Setup Block diagram



Test Data

Test Mode:	Transmitting			**Test Engineer:**			Spike Gao
Test Date:	2026-05-26			**Environment:**			Temp.: 23.2℃ Humi.: 55% Atm.:100.1kPa
Mode	**Channel**	**Conducted Output Power (dBm)**			**Result (dBm)**			**Limit (dBm)**
		LT	**NT**	**HT**	**LT**	**NT**	**HT**	
BLE 1Mbps	Low	2.756	2.683	2.61	4.256	4.183	4.11	≤ 20
	Middle	2.268	2.206	2.144	3.768	3.706	3.644	
	High	1.823	1.758	1.693	3.323	3.258	3.193	

Note:
1. The extreme operating temperature range was declared by the manufacturer.
2. The antenna gain is 1.5 dBi.

ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.2.3 - POWER SPECTRAL DENSITY

Applicable Standard

According to ETSI EN 300 328 V2.2.2 (2019-07) Clause4.3.2.3.1, this requirement applies to all types of equipment using wide band modulations other than FHSS.

The Power Spectral Density is the mean equivalent isotropically radiated power (e.i.r.p.) spectral density in a 1 MHz bandwidth during a transmission burst.

For equipment using wide band modulations other than FHSS, the maximum Power Spectral Density is limited to 10 dBm per MHz.

Test Procedure

Option 1: For equipment with continuous and non-continuous transmissions

The transmitter shall be connected to a spectrum analyser and the Power Spectral Density (PSD) as defined in clause 4.3.2.3 shall be measured and recorded.

The test procedure shall be as follows:

Step 1:

Connect the UUT to the spectrum analyser and use the following settings:

- Start Frequency: 2 400 MHz
- Stop Frequency: 2 483.5 MHz
- Resolution BW: 10 kHz
- Video BW: 30 kHz
- Sweep Points: > 8 350

NOTE: For spectrum analysers not supporting this number of sweep points, the frequency band may be segmented.

- Detector: RMS
- Trace Mode: Max Hold
- Sweep time: For non-continuous transmissions: 2 × Channel Occupancy Time × number of Sweeppoints

For continuous transmissions: 10 s; the sweep time may be increased further until a value where the sweep time has no further impact anymore on the RMS value of the signal.

NOTE: For non-continuous signals, wait for the trace to stabilize.

Save the data (trace data) set to a file.

Step 2:

For conducted measurements on smart antenna systems using either operating mode 2 or operating mode 3 (see clause 5.3.2.2), repeat the measurement for each of the transmit ports. For each sampling point (frequency domain), add up the coincident power values (in mW) for the different transmit chains and use this as the new data set.

Step 3:

Add up the values for power for all the samples in the file using the formula below.

$$P_{Sum} = \sum_{n=1}^{k} P_{sample}(n)$$

with 'k' being the total number of samples and 'n' the actual sample number

Step 4:

Normalize the individual values for power (in dBm) so that the sum is equal to the RF Output Power (e.i.r.p.) measured in clause 5.4.2 and save the corrected data. The following formulas can be used:

$$C_{Corr} = P_{Sum} - P_{e.i.r.p}$$

$$P_{Samplecorr}(n) = P_{Sample}(n) - C_{Corr}$$

with 'n' being the actual sample number

Step 5:

Starting from the first sample $P_{Samplecorr}(n)$ (lowest frequency), add up the power (in mW) of the following samples representing a 1 MHz segment and record the results for power and position (i.e. sample #1 to sample #100). This is the Power Spectral Density (e.i.r.p.) for the first 1 MHz segment which shall be recorded.

Step 6:

Shift the start point of the samples added up in step 5 by one sample and repeat the procedure in step 5 (i.e. sample #2 to sample #101).

Step 7:

Repeat step 6 until the end of the data set and record the Power Spectral Density values for each of the 1 MHz segments.

From all the recorded results, the highest value is the maximum Power Spectral Density(PSD) for the UUT. This value, which shall comply with the limit given in clause 4.3.2.3.3, shall be recorded in the test report.

Option 2: For equipment with continuous transmission capability

This option is for equipment that can be configured to operate in a continuous transmit mode (100 % DC).

Step 1:

Connect the UUT to the spectrum analyser and use the following settings:

- Centre Frequency: The centre frequency of the channel under test
- RBW: 1 MHz
- VBW: 3 MHz
- Frequency Span: At least 2 × Occupied Channel Bandwidth
- Detector Mode: Peak
- Trace Mode: Max Hold

Step 2:

When the trace is complete, find the peak value of the power envelope and record the frequency.

Step 3:

Make the following changes to the settings of the spectrum analyser:

- Centre Frequency: Equal to the frequency recorded in step 2
- Frequency Span: 3 MHz
- RBW: 1 MHz
- VBW: 3 MHz
- Sweep Time: 1 minute
- Detector Mode: RMS
- Trace Mode: Max Hold

Step 4:

Wait until trace is complete, the trace shall be captured using the "Hold" or "View" option on the spectrum analyser.

Find the peak value of the trace and place the analyser marker on this peak. This level is recorded as the highest mean power (power spectral density) D in a 1 MHz band.

Alternatively, where a spectrum analyser is equipped with a function to measure power spectral density, this function may be used to display the power spectral density D in dBm / MHz.

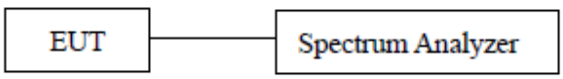
In case of conducted measurements on smart antenna systems operating in a mode with multiple transmit chains active simultaneously, the power spectral density of each transmit chain shall be measured separately to calculate the total power spectral density (value D in dBm / MHz) for the UUT.

Step 5:

The maximum Power Spectral Density (PSD) e.i.r.p. is calculated from the above measured power spectral density D, the applicable antenna assembly gain G in dBi and if applicable the beamforming gain Y in dB, according to the formula below. This value shall be recorded in the test report. If more than one antenna assembly is intended for this power setting, the gain of the antenna assembly with the highest gain shall be used.

$$PSD = D + G + Y \text{ (dBm / MHz)}$$

Test Setup Block diagram

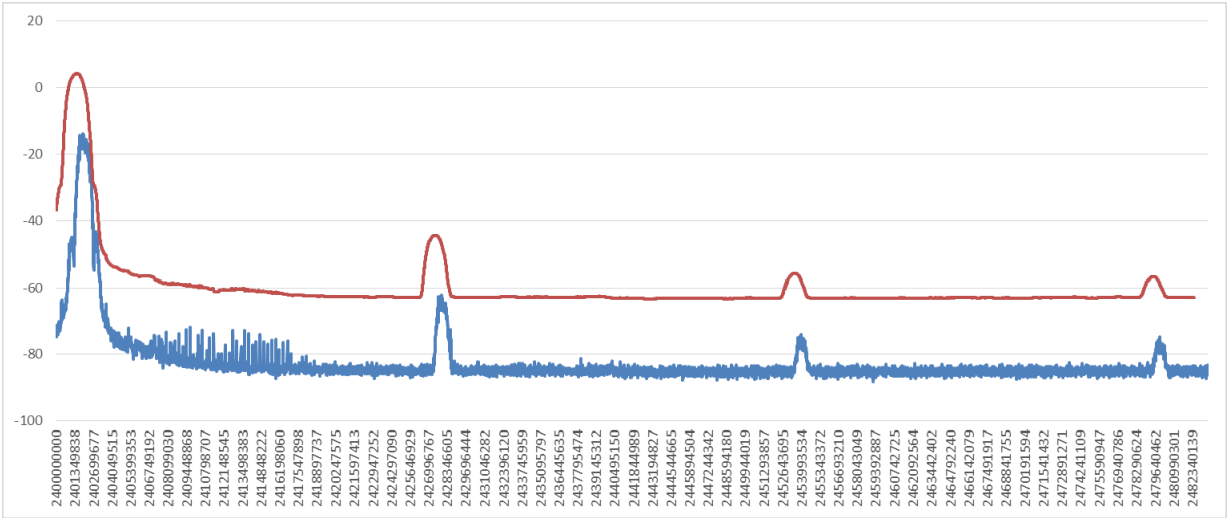


Test Data

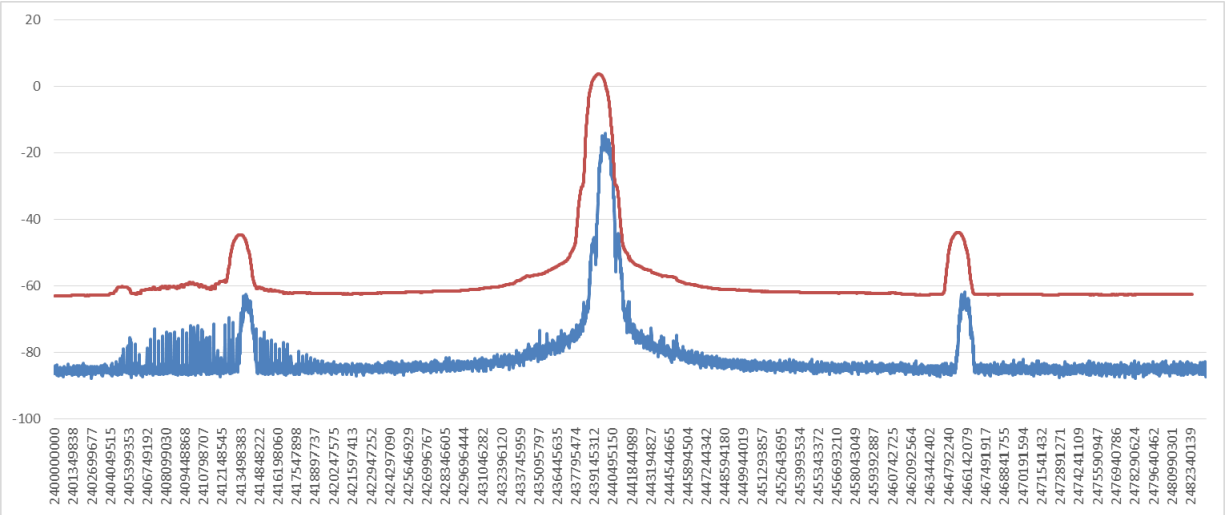
Test Mode:	Transmitting	**Test Engineer:**	Spike Gao
Test Date:	2026-05-26	**Environment:**	Temp.: 23.2℃ Humi.: 55% Atm.:100.1kPa
Mode	**Channel**	**Reading (dBm/MHz)**	**Limit (dBm/MHz)**
BLE 1Mbps	Low	4.119	≤ 10
	Middle	3.642	
	High	3.194	

Note: Antenna gain is 1.5dBi.

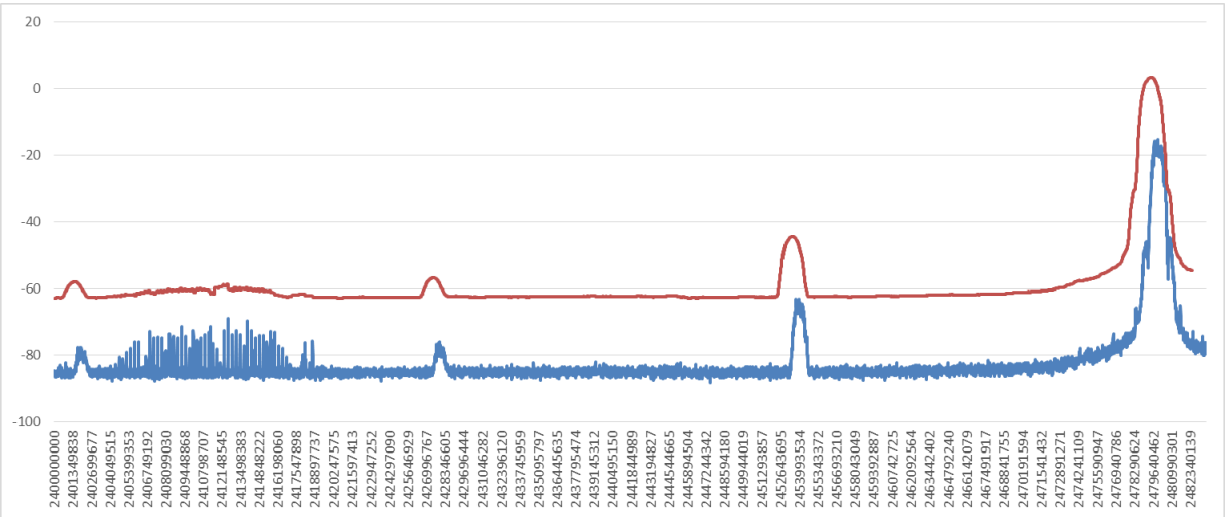
BLE 1Mbps - Low Channel: 2402 MHz



BLE 1Mbps - Middle Channel: 2440 MHz



BLE 1Mbps - High Channel: 2480 MHz



ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.2.7 – OCCUPIED CHANNEL BANDWIDTH

Applicable Standard

According to ETSI EN 300 328 V2.2.2 (2019-07) Clause4.3.2.7.2, the occupied channel bandwidth is the bandwidth that contains 99 % of the power of the signal.

Limit:

The Occupied Channel Bandwidth shall fall completely within the band given in table 1.
In addition, for non-adaptive equipment using wide band modulations other than FHSS and with e.i.r.p greater than 10 dBm, the occupied channel bandwidth shall be less than 20 MHz.

Test Procedure

The measurement procedure shall be as follows:

Step 1:

Connect the UUT to the spectrum analyser and use the following settings:

- Centre Frequency: The centre frequency of the channel under test
- Resolution BW: ~ 1 % of the span without going below 1 %
- Video BW: 3 × RBW
- Frequency Span: 2 × Nominal Channel Bandwidth
- Detector Mode: RMS
- Trace Mode: Max Hold
- Sweep time: 1 s

Step 2:

Wait for the trace to stabilize.
Find the peak value of the trace and place the analyser marker on this peak.

Step 3:

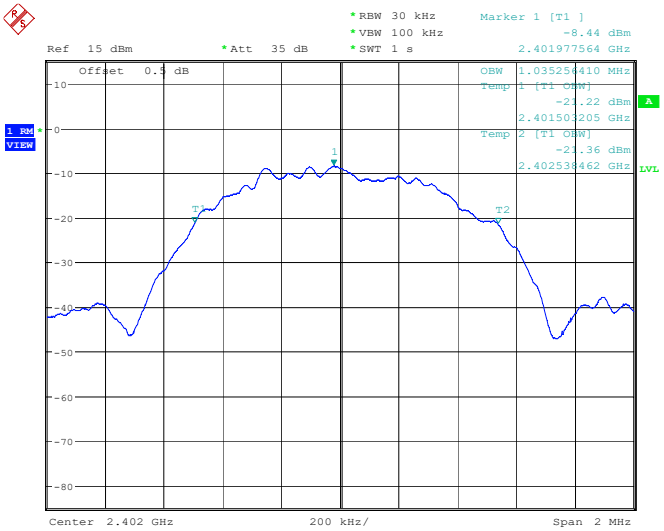
Use the 99 % bandwidth function of the spectrum analyser to measure the Occupied Channel Bandwidth of the UUT. This value shall be recorded.

NOTE: Make sure that the power envelope is sufficiently above the noise floor of the analyser to avoid the noise signals left and right from the power envelope being taken into account by this measurement.

Test Data

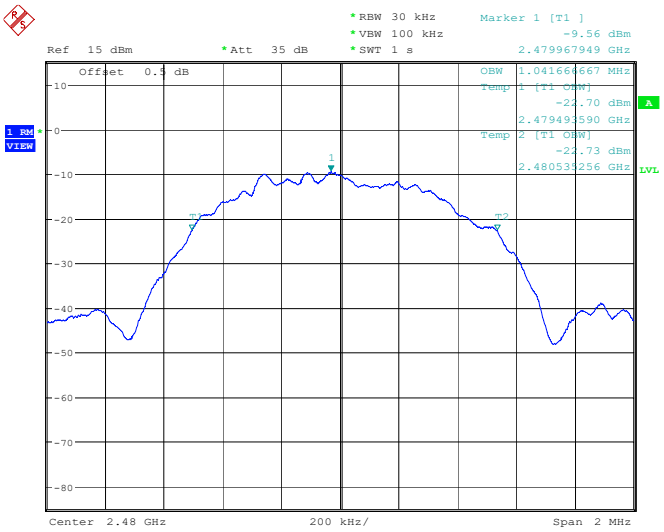
Test Mode:	Transmitting	**Test Engineer:**	Spike Gao
Test Date:	2026-05-26	**Environment:**	Temp.: 23.2℃ Humi.: 55% Atm.:100.1kPa
Mode	**Channel**	**Frequency (MHz)**	**Result (MHz)**
BLE 1Mbps	Low	2402	1.035
	High	2480	1.042

BLE 1Mbps - Low Channel



ProjectNo.:2607T13127E-RF Tester:Spike Gao
Date: 26.MAY.2026 10:57:39

BLE 1Mbps - High Channel



ProjectNo.:2607T13127E-RF Tester:Spike Gao
Date: 26.MAY.2026 10:58:14

ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.2.8 – TRANSMITTER UNWANTED EMISSION IN THE OUT-OF-BAND DOMAIN

Applicable Standard

According to ETSI EN 300 328 V2.2.2 (2019-07) Clause4.3.2.8.2, Transmitter unwanted emissions in the out-of-band domain are emissions when the equipment is in Transmit mode, on frequencies immediately outside the necessary bandwidth which results from the modulation process, but excluding spurious emissions.

Limit:

The transmitter unwanted emissions in the out-of-band domain but outside the allocated band, shall not exceed the values provided by the mask in figure 3.

Within the band specified in table 1, the Out-of-band emissions are fulfilled by compliance with the Occupied Channel Bandwidth requirement in clause 4.3.2.7.

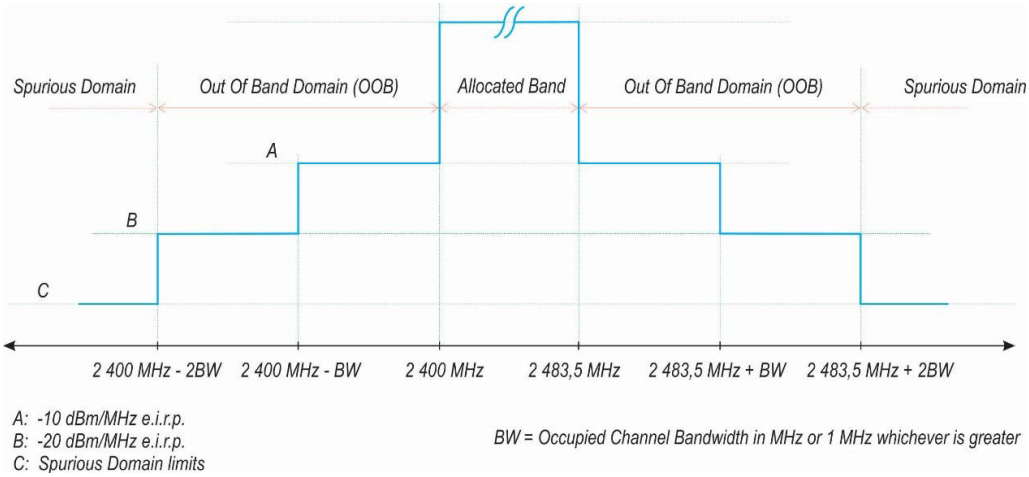


Figure 3: Transmit mask

Test Procedure

The measurement procedure shall be as follows:

Step 1:
• Connect the UUT to the spectrum analyser and use the following settings:
- Measurement Mode: Time Domain Power
- Centre Frequency: 2 484 MHz
- Span: Zero Span
- Resolution BW: 1 MHz
- Filter mode: Channel filter
- Video BW: 3 MHz
- Detector Mode: RMS
- Trace Mode: Max Hold
- Sweep Mode: Single Sweep
- Sweep Points: Sweep time [µs] / (1 µs) with a maximum of 30 000
- Trigger Mode: Video
- Sweep Time: > 120 % of the duration of the longest burst detected during the measurement of the RF Output Power

Step 2 (segment 2 483,5 MHz to 2 483,5 MHz + BW):

• The measurement shall be performed and repeated while the trigger level is increased until no triggering takes place.

• For FHSS equipment operating in a normal hopping mode, the different hops will result in signal bursts with different power levels. In this case the burst with the highest power level shall be selected.

• Set a window (start and stop lines) to match with the start and end of the burst and in which the RMS power shall be measured using the Time Domain Power function.

• Select RMS power to be measured within the selected window and note the result which is the RMS power within this 1 MHz segment (2 483,5 MHz to 2 484,5 MHz). Compare this value with the applicable limit provided by the mask.

• Increase the centre frequency in steps of 1 MHz and repeat this measurement for every 1 MHz segment within the range 2 483,5 MHz to 2 483,5 MHz + BW. The centre frequency of the last 1 MHz segment shall be set to 2 483,5 MHz + BW - 0,5 MHz (which means this may partly overlap with the previous 1 MHz segment).

Step 3 (segment 2 483,5 MHz + BW to 2 483,5 MHz + 2BW):

• Change the centre frequency of the analyser to 2 484 MHz + BW and perform the measurement for the first 1 MHz segment within range 2 483,5 MHz + BW to 2 483,5 MHz + 2BW. Increase the centre frequency in 1 MHz steps and repeat the measurements to cover this whole range. The centre frequency of the last 1 MHz segment shall be set to 2 483,5 MHz + 2 BW - 0,5 MHz (which means this may partly overlap with the previous 1 MHz segment).

Step 4 (segment 2 400 MHz - BW to 2 400 MHz):

• Change the centre frequency of the analyser to 2 399,5 MHz and perform the measurement for the first 1 MHz segment within range 2 400 MHz - BW to 2 400 MHz Reduce the centre frequency in 1 MHz steps and repeat the measurements to cover this whole range. The centre frequency of the last 1 MHz segment shall be set to 2 400 MHz - BW + 0,5 MHz (which means this may partly overlap with the previous 1 MHz segment).

Step 5 (segment 2 400 MHz - 2BW to 2 400 MHz - BW):

• Change the centre frequency of the analyser to 2 399,5 MHz - BW and perform the measurement for the first 1 MHz segment within range 2 400 MHz - 2BW to 2 400 MHz - BW. Reduce the centre frequency in 1 MHz steps and repeat the measurements to cover this whole range. The centre frequency of the last 1 MHz segment shall be set to 2 400 MHz - 2BW + 0,5 MHz (which means this may partly overlap with the previous 1 MHz segment).

Step 6:

• In case of conducted measurements on equipment with a single transmit chain, the declared antenna assembly gain G in dBi shall be added to the results for each of the 1 MHz segments and compared with the limits provided by the mask given in figure 1 or figure 3. If more than one antenna assembly is intended for this power setting, the antenna with the highest gain shall be considered.

• In case of conducted measurements on smart antenna systems (equipment with multiple transmit chains), the measurements need to be repeated for each of the active transmit chains. The declared antenna assembly gain G in dBi for a single antenna shall be added to these results. If more than one antenna assembly is intended for this power setting, the antenna with the highest gain shall be considered. Comparison with the applicable limits shall be done using any of the options given below:

- Option 1: the results for each of the transmit chains for the corresponding 1 MHz segments shall be added. The additional beamforming gain Y in dB shall be added as well and the resulting values compared with the limits provided by the mask given in figure 1 or figure 3.

- Option 2: the limits provided by the mask given in figure 1 or figure 3 shall be reduced by 10 × log10(Ach) and the additional beamforming gain Y in dB. The results for each of the transmit chains shall be individually compared with these reduced limits.

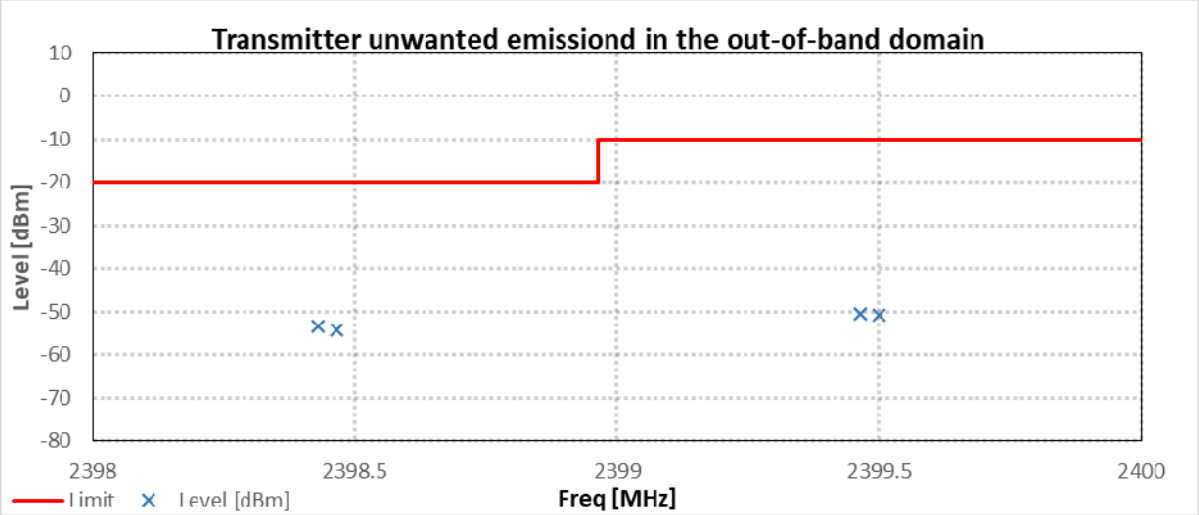
NOTE: A_{ch} refers to the number of active transmit chains.

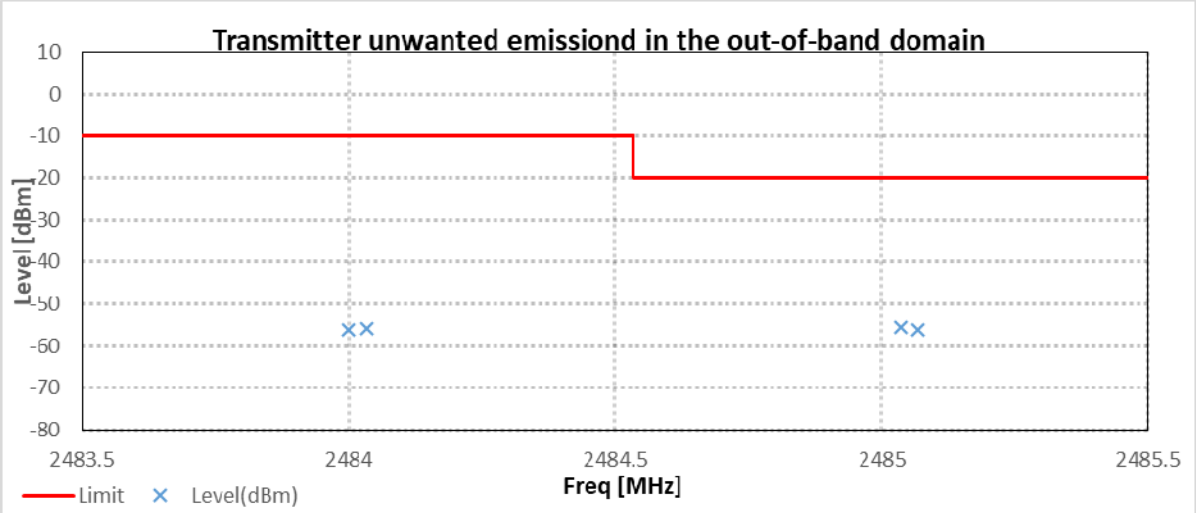
It shall be recorded whether the equipment complies with the mask provided in figure 1 or figure 3.

Test Data

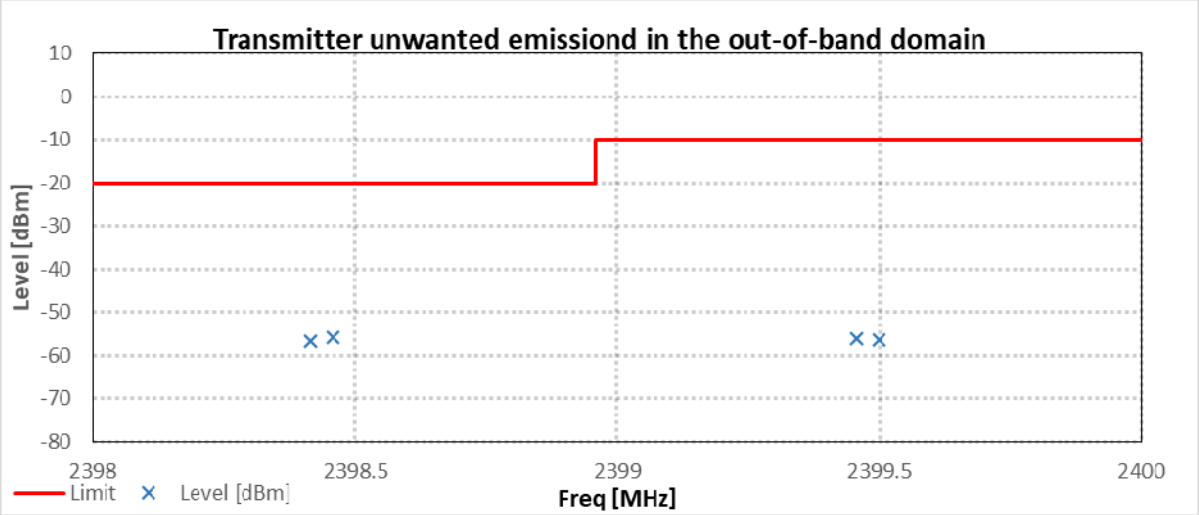
Test Mode:	Transmitting	**Test Engineer:**	Spike Gao
Test Date:	2026-05-26	**Environment:**	Temp.: 23.2℃ Humi.: 55% Atm.: 100.1kPa

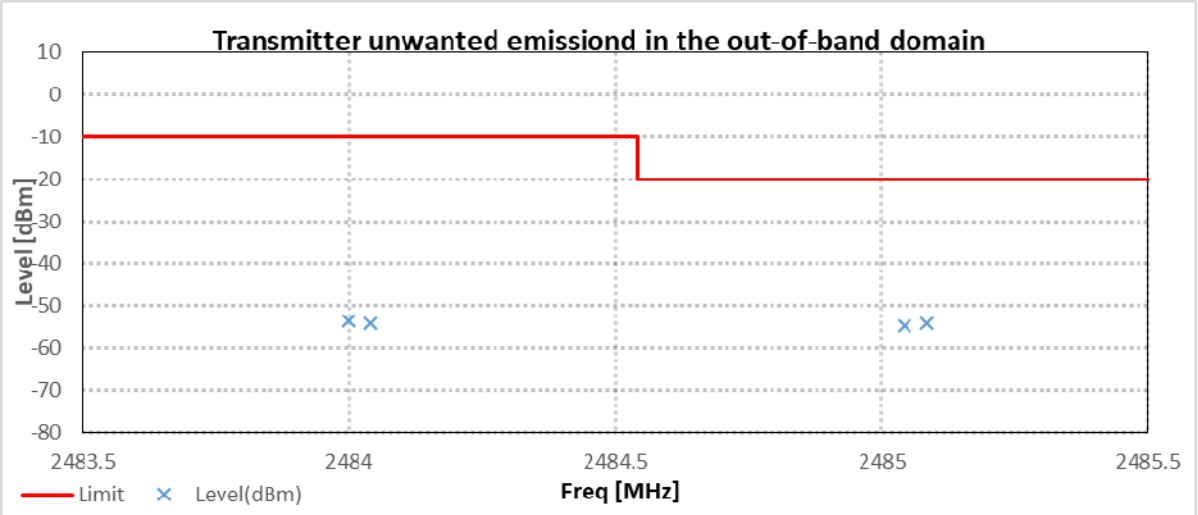
BLE 1Mps Mode: 2402 MHz





BLE 1Mps Mode: 2480 MHz





ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.2.9 – TRANSMITTER UNWANTED EMISSION IN THE SPURIOUS DOMAIN

Applicable Standard

Transmitter unwanted emissions in the spurious domain are emissions outside the allocated band and outside the Out-of-band Domain as indicated in figure 3 when the equipment is in Transmit mode.

The transmitter unwanted emissions in the spurious domain shall not exceed the values given in the following table.

Transmitter limits for spurious emissions

Frequency Range	Maximum power e.r.p (≤ 1 GHz) e.i.r.p (> 1 GHz)	Bandwidth
30 MHz to 47 MHz	-36 dBm	100 kHz
47 MHz to 74 MHz	-54 dBm	100 kHz
74 MHz to 87,5 MHz	-36 dBm	100 kHz
87,5 MHz to 118 MHz	-54 dBm	100 kHz
118 MHz to 174 MHz	-36 dBm	100 kHz
174 MHz to 230 MHz	-54 dBm	100 kHz
230 MHz to 470 MHz	-36 dBm	100 kHz
470 MHz to 694 MHz	-54 dBm	100 kHz
694 MHz to 1 GHz	-36 dBm	100 kHz
1 GHz to 12,75 GHz	-30 dBm	1MHz

NOTE: In case of equipment with antenna connectors, these limits apply to emissions at the antenna port (conducted). For emissions radiated by the cabinet or emissions radiated by integral antenna equipment (without antenna connectors), these limits are e.r.p. for emissions up to 1 GHz and as e.i.r.p. for emissions above 1 GHz.

Test Procedure

Pre-scan:

The procedure in step 1 to step 4 below shall be used to identify potential unwanted emissions of the UUT.

Step 1:

The sensitivity of the spectrum analyser should be such that the noise floor is at least 12 dB below the limits given in table 4 or table 12.

Step 2:

The emissions over the range 30 MHz to 1 000 MHz shall be identified.

Spectrum analyser settings:

- Resolution bandwidth: 100 kHz
- Video bandwidth: 300 kHz
- Filter type: 3 dB (Gaussian)
- Detector mode: Peak
- Trace Mode: Max Hold
- Sweep Points: ≥ 19 400; for spectrum analysers not supporting this high number of sweep points, the frequency band may be segmented.
- Sweep time: For non continuous transmissions (duty cycle less than 100 %), the sweep time shall be sufficiently long, such that for each 100 kHz frequency step, the measurement time is greater than two transmissions of the UUT, on any channel.

Allow the trace to stabilize. Any emissions identified during the sweeps above and that fall within the 6 dB range below the applicable limit or above, shall be individually measured using the procedure in clause 5.4.9.2.1.3 and compared tothe limits given in table 4 or table 12.

Step 3:

The emissions over the range 1 GHz to 12,75 GHz shall be identified.

Spectrum analyser settings:

- Resolution bandwidth: 1 MHz
- Video bandwidth: 3 MHz
- Filter type: 3 dB (Gaussian)
- Detector mode: Peak
- Trace Mode: Max Hold
- Sweep Points: ≥ 23 500; for spectrum analysers not supporting this high number of sweep points, the frequency band may be segmented
- Sweep time: For non continuous transmissions (duty cycle less than 100 %), the sweep time shall be sufficiently long, such that for each 1 MHz frequency step, the measurement time is greater than two transmissions of the UUT, on any channel.

Allow the trace to stabilize. Any emissions identified during the sweeps above that fall within the 6 dB range below theapplicable limit or above, shall be individually measured using the procedure in clause 5.4.9.2.1.3 and compared to thelimits given in table 4 or table 12.

Step 4:

- In case of conducted measurements on smart antenna systems (equipment with multiple receive chains), step 2 and step 3 need to be repeated for each of the active receive chains (A_{ch}).The limits used to identify emissions during this pre-scan need to be reduced by $10 \times \log10 (A_{ch})$.

Measurement of the emissions identified during the pre-scan:

The procedure in step 1 to step 4 below shall be used to accurately measure the individual unwanted emissions identified during the pre-scan measurements above. This method assumes the spectrum analyser has a Time Domain Power function.

Step 1:

The level of the emissions shall be measured using the following spectrum analyser settings:

- Measurement Mode: Time Domain Power
- Centre Frequency: Frequency of the emission identified during the pre-scan
- Resolution Bandwidth: 100 kHz (< 1 GHz) / 1 MHz (> 1 GHz)
- Video Bandwidth: 300 kHz (< 1 GHz) / 3 MHz (> 1 GHz)
- Frequency Span: Zero Span
- Sweep Mode: Single Sweep
- Sweep time: > 120 % of the duration of the longest burst detected during the measurement of the RF Output Power
- Sweep points: Sweep time [µs] / (1 µs) with a maximum of 30 000
- Trigger: Video (burst signals) or Manual (continuous signals)
- Detector: RMS

Step 2:

Set a window where the start and stop indicators match the start and end of the burst with the highest level and record the value of the power measured within this window. If the spurious emission to be measured is a continuous transmission, the measurement window shall be set to match the start and stop times of the sweep.

Step 3:

In case of conducted measurements on smart antenna systems (equipment with multiple transmit chains), step 2 needs to be repeated for each of the active transmit chains (A_{ch}).

Sum the measured power (within the observed window) for each of the active transmit chains.

Step 4:

The value defined in step 3 shall be compared to the limits defined in table 4 or table 12.

Test Data

Test result: Compliant

Test Mode:	Transmitting	**Test Engineer:**	Wlif Wu
Test Date:	2026-06-24~2026-06-26	**Environment:**	Temp.: 22.7~23.2℃ Humi.: 59~60% Atm.: 100.1kPa

30 MHz - 12.75 GHz								
Frequency (MHz)	**Polar (H/V)**	**Receiver Reading (dBμV)**	**Substituted Method**			**Absolute Level (dBm)**	**Limit (dBm)**	**Margin (dB)**
			Substituted Level (dBm)	**Antenna Gain (dBd/dBi)**	**Cable Loss (dB)**			
BLE 1Mbps - low channel (2402 MHz)								
517.86	H	30.75	-72.15	0.00	0.37	-72.52	-54.00	18.52
173.61	V	34.19	-75.27	0.00	0.24	-75.51	-36.00	39.51
261.01	H	29.80	-67.57	0.00	0.28	-67.85	-36.00	31.85
250.53	V	29.75	-64.04	0.00	0.28	-64.32	-36.00	28.32
4804.000	H	55.69	-49.90	11.80	2.69	-40.79	-30.00	10.79
4804.000	V	51.95	-53.94	11.80	2.69	-44.83	-30.00	14.83
7206.000	H	46.48	-54.46	9.00	3.29	-48.75	-30.00	18.75
7206.000	V	46.61	-55.05	9.00	3.29	-49.34	-30.00	19.34
BLE 1Mbps - high channel (2480 MHz)								
697.22	H	31.37	-63.90	0.00	0.40	-64.30	-36.00	28.30
172.06	V	34.81	-74.77	0.00	0.24	-75.01	-36.00	39.01
576.45	H	29.10	-74.71	0.00	0.38	-75.09	-54.00	21.09
110.07	V	28.20	-85.45	0.00	0.21	-85.66	-54.00	31.66
4960.000	H	64.82	-40.84	10.68	2.72	-32.88	-30.00	2.88
4960.000	V	56.64	-49.49	10.68	2.72	-41.53	-30.00	11.53
7440.000	H	49.09	-51.54	7.86	3.33	-47.01	-30.00	17.01
7440.000	V	47.88	-53.80	7.86	3.33	-49.27	-30.00	19.27

Note: The unit of antenna gain is dBd for frequency below 1GHz and is dBi for frequency above 1GHz.
Absolute Level = Submitted Level - Cable loss + Antenna Gain
Margin = Limit- Absolute Level

ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.2.10 – RECEIVER SPURIOUS EMISSIONS

Applicable Standard

According to ETSI EN 300 328 V2.2.2 (2019-07) Clause4.3.2.10.2, the receiver spurious emissions are emissions at any frequency when the equipment is in receive mode.

The spurious emissions of the receiver shall not exceed the values given in the following table

Frequency range	Maximum power, e.r.p.	Measurement bandwidth
30 MHz to 1 GHz	-57 dBm	100 kHz
1 GHz to 12,75 GHz	-47 dBm	1 MHz

NOTE: In case of equipment with antenna connectors, these limits apply to emissions at the antenna port (conducted). For emissions radiated by the cabinet or for emissions radiated by integral antenna equipment (without antenna connectors), these limits are e.r.p. for emissions up to 1 GHz and e.i.r.p. for emissions above 1 GHz.

Test Procedure

Pre-scan:

The procedure in step 1 to step 4 below shall be used to identify potential unwanted emissions of the UUT.

Step 1:

The sensitivity of the spectrum analyser should be such that the noise floor is at least 12 dB below the limits given in table 5 or table 13.

Step 2:

The emissions over the range 30 MHz to 1 000 MHz shall be identified.
Spectrum analyser settings:

- Resolution bandwidth: 100 kHz
- Video bandwidth: 300 kHz
- Filter type: 3 dB (Gaussian)
- Detector mode: Peak
- Trace Mode: Max Hold
- Sweep Points: ≥ 19 400
- Sweep time: Auto

Note: Wait for the trace to stabilize. Any emissions identified during the sweeps above and that fall within the 6 dB range below the applicable limit or above, shall be individually measured using the procedure in clause 5.4.10.2.1.3 and compared to the limits given in table 5 or table 13.

Step 3:

The emissions over the range 1 GHz to 12,75 GHz shall be identified.

Spectrum analyser settings:

- Resolution bandwidth: 1 MHz

- Video bandwidth: 3 MHz
- Filter type: 3 dB (Gaussian)
- Detector mode: Peak
- Trace Mode: Max Hold
- Sweep Points: ≥ 23 500; for spectrum analysers not supporting this high number of sweep points, the frequency band may be segmented
- Sweep time: Auto

Wait for the trace to stabilize. Any emissions identified during the sweeps above that fall within the 6 dB range below the applicable limit or above, shall be individually measured using the procedure in clause 5.4.10.2.1.3 and compared to the limits given in table 5 or table 13.

Frequency Hopping equipment may generate a block (or several blocks) of spurious emissions anywhere within the spurious domain. If this is the case, only the highest peak of each block of emissions shall be measured using the procedure in clause 5.4.10.2.1.3.

Step 4:

In case of conducted measurements on smart antenna systems (equipment with multiple receive chains), step 2 and step 3 need to be repeated for each of the active receive chains A_{ch}The limits used to identify emissions during this pre-scan need to be reduced by $10 \times \log10\ A_{ch}$.

Measurement of the emissions identified during the pre-scan:

The procedure in step 1 to step 4 below shall be used to accurately measure the individual unwanted emissions identified during the pre-scan measurements above. This method assumes the spectrum analyser has a Time Domain Power function.

Step 1:

The level of the emissions shall be measured using the following spectrum analyser settings:

- Measurement Mode: Time Domain Power
- Centre Frequency: Frequency of the emission identified during the pre-scan
- Resolution Bandwidth: 100 kHz (< 1 GHz) / 1 MHz (> 1 GHz)
- Video Bandwidth: 300 kHz (< 1 GHz) / 3 MHz (> 1 GHz)
- Frequency Span: Zero Span
- Sweep mode: Single Sweep
- Sweep time: 30 ms
- Sweep points: ≥ 30 000
- Trigger: Video (for burst signals) or Manual (for continuous signals)
- Detector: RMS

Step 2:

Set a window where the start and stop indicators match the start and end of the burst with the highest level and record the value of the power measured within this window. If the spurious emission to be measured is a continuous transmission, the measurement window shall be set to the start and stop times of the sweep.

Step 3:

In case of conducted measurements on smart antenna systems (equipment with multiple receive chains), step 2 needs to be repeated for each of the active receive chains A_{ch}.

Sum the measured power (within the observed window) for each of the active receive chains.

Step 4:

The value defined in step 3 shall be compared to the limits defined in table 5 and table 13.

Test Data

Test result: Compliant.

Test Mode:	Receiving	**Test Engineer:**	Wlif Wu
Test Date:	2026-06-24~2026-06-26	**Environment:**	Temp.: 22.7~23.2℃ Humi.: 59~60% Atm.: 100.1kPa

30 MHz - 12.75 GHz								
Frequency (MHz)	**Polar (H/V)**	**Receiver Reading (dBμV)**	**Substituted Method**			**Absolute Level (dBm)**	**Limit (dBm)**	**Margin (dB)**
			Substituted Level (dBm)	**Antenna Gain (dBd/dBi)**	**Cable Loss (dB)**			
BLE 1Mbps - low channel (2402 MHz)								
732.14	H	30.28	-67.83	0.00	0.41	-68.24	-57.00	11.24
173.51	V	32.14	-77.33	0.00	0.24	-77.57	-57.00	20.57
635.33	H	29.93	-69.11	0.00	0.39	-69.50	-57.00	12.50
307.47	V	31.68	-76.67	0.00	0.29	-76.96	-57.00	19.96
1816.625	H	51.51	-60.35	6.70	1.71	-55.36	-47.00	8.36
1816.625	V	57.78	-54.37	6.70	1.71	-49.38	-47.00	2.38
BLE 1Mbps - high channel (2480 MHz)								
655.12	H	30.48	-66.81	0.00	0.39	-67.20	-57.00	10.20
173.61	V	34.05	-75.41	0.00	0.24	-75.65	-57.00	18.65
523.68	H	29.75	-73.55	0.00	0.37	-73.92	-57.00	16.92
106.39	V	29.68	-84.19	0.00	0.20	-84.39	-57.00	27.39
1816.625	H	52.55	-59.31	6.70	1.71	-54.32	-47.00	7.32
1817.800	V	54.71	-57.43	6.70	1.71	-52.44	-47.00	5.44

Note: The unit of antenna gain is dBd for frequency below 1GHz and is dBi for frequency above 1GHz.
Absolute Level = Submitted Level - Cable loss + Antenna Gain
Margin = Limit- Absolute Level

ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.2.11 – RECEIVER BLOCKING

Applicable Standard

Receiver blocking is a measure of the ability of the equipment to receive a wanted signal on its operating channel without exceeding a given degradation in the presence of an unwanted signal (blocking signal) at frequencies other than those of the operating band.

While maintaining the minimum performance criteria as defined in clause 4.3.2.11.3, the blocking levels at specified frequency offsets shall be equal to or greater than the limits defined for the applicable receiver category provided in table 14, table 15 or table 16.

Table 14: Receiver Blocking parameters for Receiver Category 1 equipment

Wanted signal mean power from companion device (dBm) (see notes 1 and 4)	Blocking signal frequency (MHz)	Blocking signal power (dBm) (see note 4)	Type of blocking signal
(-133 dBm + 10 × $\log_{10}$(OCBW)) or -68 dBm whichever is less (see note 2)	2 380 2 504	-34	CW
(-139 dBm + 10 × $\log_{10}$(OCBW)) or -74 dBm whichever is less (see note 3)	2 300 2 330 2 360 2 524 2 584 2 674		

NOTE 1: OCBW is in Hz.

NOTE 2: In case of radiated measurements using a companion device and the level of the wanted signal from the companion device cannot be determined, a relative test may be performed using a wanted signal up to P_{min} + 26 dB where P_{min} is the minimum level of wanted signal required to meet the minimum performance criteria as defined in clause 4.3.1.12.3 in the absence of any blocking signal.

NOTE 3: In case of radiated measurements using a companion device and the level of the wanted signal from the companion device cannot be determined, a relative test may be performed using a wanted signal up to P_{min} + 20 dB where P_{min} is the minimum level of wanted signal required to meet the minimum performance criteria as defined in clause 4.3.1.12.3 in the absence of any blocking signal.

NOTE 4: The level specified is the level at the UUT receiver input assuming a 0 dBi antenna assembly gain. In case of conducted measurements, this level has to be corrected for the (in-band) antenna assembly gain (G). In case of radiated measurements, this level is equivalent to a power flux density (PFD) in front of the UUT antenna with the UUT being configured/positioned as recorded in clause 5.4.3.2.2.

Table 15: Receiver Blocking parameters receiver Category 2 equipment

Wanted signal mean power from companion device (dBm) (see notes 1 and 3)	Blocking signal frequency (MHz)	Blocking signal power (dBm) (see note 3)	Type of blocking signal
(-139 dBm + 10 × $\log_{10}$(OCBW) + 10 dB) or (-74 dBm + 10 dB) whichever is less (see note 2)	2 380 2 504 2 300 2 584	-34	CW
NOTE 1: OCBW is in Hz. NOTE 2: In case of radiated measurements using a companion device and the level of the wanted signal from the companion device cannot be determined, a relative test may be performed using a wanted signal up to P_{min} + 26 dB where P_{min} is the minimum level of wanted signal required to meet the minimum performance criteria as defined in clause 4.3.1.12.3 in the absence of any blocking signal. NOTE 3: The level specified is the level at the UUT receiver input assuming a 0 dBi antenna assembly gain. In case of conducted measurements, this level has to be corrected for the (in-band) antenna assembly gain (G). In case of radiated measurements, this level is equivalent to a power flux density (PFD) in front of the UUT antenna with the UUT being configured/positioned as recorded in clause 5.4.3.2.2.			

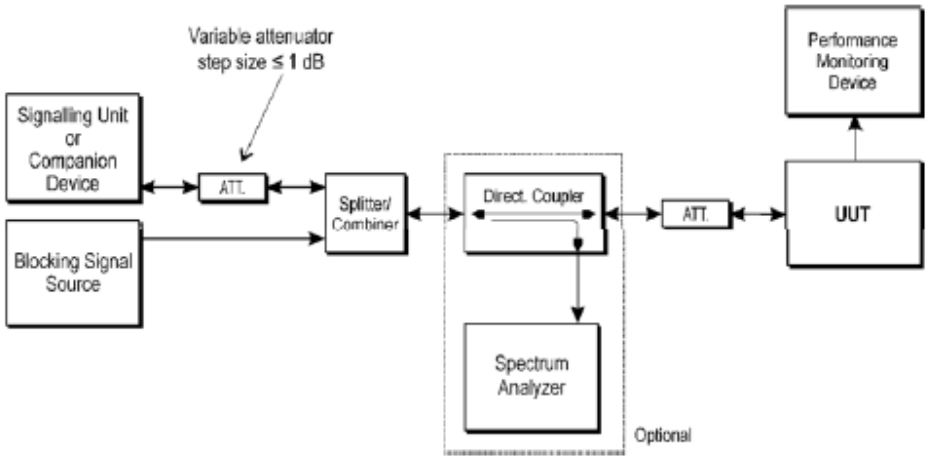
Table 16: Receiver Blocking parameters receiver Category 3 equipment

Wanted signal mean power from companion device (dBm) (see notes 1 and 3)	Blocking signal frequency (MHz)	Blocking signal power (dBm) (see note 3)	Type of blocking signal
(-139 dBm + 10 × $\log_{10}$(OCBW) + 20 dB) or (-74 dBm + 20 dB) whichever is less (see note 2)	2 380 2 504 2 300 2 584	-34	CW
NOTE 1: OCBW is in Hz. NOTE 2: In case of radiated measurements using a companion device and the level of the wanted signal from the companion device cannot be determined, a relative test may be performed using a wanted signal up to P_{min} + 30 dB where P_{min} is the minimum level of wanted signal required to meet the minimum performance criteria as defined in clause 4.3.1.12.3 in the absence of any blocking signal. NOTE 3: The level specified is the level at the UUT receiver input assuming a 0 dBi antenna assembly gain. In case of conducted measurements, this level has to be corrected for the (in-band) antenna assembly gain (G). In case of radiated measurements, this level is equivalent to a power flux density (PFD) in front of the UUT antenna with the UUT being configured/positioned as recorded in clause 5.4.3.2.2.			

Test Procedure

According to ETSI EN 300 328 V2.2.2 (2019-07) Clause5.4.11.2

Test Setup Block diagram



Test data

Test Result: Compliant. Please refer to following tables.

Test Mode:	Receiving		**Test Engineer:**	Spike Gao	
Test Date:	2026-05-26		**Environment:**	Temp.: 23.2℃ Humi.: 55% Atm.: 100.1kPa	
Mode	**Receiver Category**	**Channel**	**Blocking signal frequency (MHz)**	**PER (%)**	**Limit (%)**
BLE 1Mbps	2	Low	2300	2.16	≤ 10
			2380	2.19	
		High	2504	2.13	
			2584	1.73	

Note: PER was monitored by CMW500.

EXHIBIT A - E.2 INFORMATION AS REQUIRED BY ETSI EN 300 328 V2.2.2, Clause 5.4.1

In accordance with EN 300 328, clause 5.4.1, the following information is provided by the manufacturer.

a) The type of modulation used by the equipment:

☐FHSS
☒other forms of modulation

b) In case of FHSS:

In case of non-Adaptive FHSS equipment:
The number of Hopping Frequencies:_____.

In case of Adaptive FHSS equipment:
The maximum number of Hopping Frequencies:_____;
The minimum number of Hopping Frequencies: _____;

The (average) Dwell Time: _____;

c) Adaptive / non-adaptive equipment:

☐non-adaptive Equipment
☒adaptive Equipment without the possibility to switch to a non-adaptive mode
☐adaptive Equipment which can also operate in a non-adaptive mode

d) In case of adaptive equipment:

The maximum Channel Occupancy Time implemented by the equipment: _____ms

☐The equipment has implemented an LBT mechanism

In case of non-FHSS equipment:

☐The equipment is Frame Based equipment
☒The equipment is Load Based equipment
☐The equipment can switch dynamically between Frame Based and Load Based equipment

The CCA time implemented by the equipment: _____μs

☐The equipment has implemented a DAA mechanism
☐The equipment can operate in more than one adaptive mode

e) In case of non-adaptive Equipment:

The maximum RF Output Power (e.i.r.p.): _____dBm
The maximum (corresponding) Duty Cycle: _____ %

Equipment with dynamic behaviour, that behaviour is described here. (e.g. the different combinations of duty cycle and corresponding power levels to be declared):

f) The worst case operational mode for each of the following tests:

RF Output Power: 4.256dBm ;
Power Spectral Density 4.119 dBm/MHz ;
Duty cycle, Tx-Sequence, Tx-gap N/A ;
Accumulated Transmit time, Frequency Occupation & Hopping Sequence (only for FHSS equipment) N/A ;
Hopping Frequency Separation (only for FHSS equipment) N/A ;
Medium Utilisation N/A ;
Adaptivity Not Applicable ;
Receiver Blocking Compliant ;
Nominal Channel Bandwidth 1 MHz ;
Transmitter unwanted emissions in the OOB domain -53.32dBm/MHz ;
Transmitter unwanted emissions in the spurious domain -32.88 dBm ;
Receiver spurious emissions -49.38 dBm ;

g) The different transmit operating modes (tick all that apply):

☒Operating mode 1: Single Antenna Equipment
☒Equipment with only one antenna
☐Equipment with two diversity antennas but only one antenna active at any moment in time
☐Smart Antenna Systems with 2 or more antennas, but operating in a (legacy) mode where only 1 antenna is used. (e.g. IEEE 802.11™ legacy mode in smart antenna systems)

☐Operating mode 2: Smart Antenna Systems - Multiple Antennas without beam forming
☐Single spatial stream / Standard throughput / (e.g. IEEE 802.11™ legacy mode)
☐High Throughput (> 1 spatial stream) using Nominal Channel Bandwidth 1
☐High Throughput (> 1 spatial stream) using Nominal Channel Bandwidth 2
Note: Add more lines if more channel bandwidths are supported.

☐Operating mode 3: Smart Antenna Systems - Multiple Antennas with beam forming
☐Single spatial stream / Standard throughput (e.g. IEEE 802.11™ legacy mode)
☐High Throughput (> 1 spatial stream) using Nominal Channel Bandwidth 1
☐High Throughput (> 1 spatial stream) using Nominal Channel Bandwidth 2
Note: Add more lines if more channel bandwidths are supported.

h) In case of Smart Antenna Systems:

The number of Receive chains: ;
The number of Transmit chains: ;

☐symmetrical power distribution
☐asymmetrical power distribution

In case of beam forming, the maximum (additional) beam forming gain: N/A ;

Note: The additional beam forming gain does not include the basic gain of a single antenna.

i) Operating Frequency Range(s) of the equipment:

Operating Frequency Range 1: 2402 MHz to 2480 MHz
Operating Frequency Range 2: _____ MHz to _____ MHz

Note: Add more lines if more Frequency Ranges are supported.

j) Nominal Channel Bandwidth(s):

Nominal Channel Bandwidth 1: 1.0 MHz
Nominal Channel Bandwidth 2: _____MHz
Nominal Channel Bandwidth 3: _____ MHz
Nominal Channel Bandwidth 4: _____ MHz

Note: Add more lines if more channel bandwidths are supported.

k) Type of Equipment (stand-alone, combined, plug-in radio device, etc.):

☒Stand-alone
☐Combined Equipment
☐Plug-in radio device
☐Other _____;

l) The normal and extreme operating conditions that apply to the equipment:

Normal operating conditions (if applicable):

Operating temperature: +25 ° C

Other (please specify if applicable):

Extreme operating conditions:

Operating temperature range: Minimum: -40 ° C Maximum: +85 ° C

Other (please specify if applicable): …… Minimum: …… Maximum ……

Details provided are for the: ☒stand-alone equipment
☐combined (or host) equipment
☐test jig

m) The intended combination(s) of the radio equipment power settings and one or more antenna assemblies and their corresponding e.i.r.p levels:

Antenna Type:

☒Integral Antenna (information to be provided in case of conducted measurements)

Antenna Gain: 1.5dBi

If applicable, additional beamforming gain (excluding basic antenna gain):_____ dB

☐Temporary RF connector provided
☐No temporary RF connector provided
☐Dedicated Antennas (equipment with antenna connector)

☐Single power level with corresponding antenna(s)
☐Multiple power settings and corresponding antenna(s)

Number of different Power Levels: .....

Power Level 1: ..... dBm
Power Level 2: ..... dBm
Power Level 3: ..... dBm

Note 1: Add more lines in case the equipment has more power levels.
Note 2: These power levels are conducted power levels (at antenna connector).

For each of the Power Levels, provide the intended antenna assemblies, their corresponding gains (G) and the resulting e.i.r.p. levels also taking into account the beamforming gain (Y) if applicable

Power Level 1: ..... dBm

Number of antenna assemblies provided for this power level: .....

Assembly #	Gain (dBi)	e.i.r.p.(dBm)	Part number or model name
1			
2			
3			
4			

NOTE 3: Add more rows in case more antenna assemblies are supported for this power level.

Power Level 2: ..... dBm

Number of antenna assemblies provided for this power level: .....

Assembly #	Gain (dBi)	e.i.r.p.(dBm)	Part number or model name
1			
2			
3			
4			

NOTE 4: Add more rows in case more antenna assemblies are supported for this power level.

Power Level 3: ..... dBm

Number of antenna assemblies provided for this power level: .....

Assembly #	Gain (dBi)	e.i.r.p.(dBm)	Part number or model name
1			
2			
3			
4			

NOTE 5: Add more rows in case more antenna assemblies are supported for this power level.

n) The nominal voltages of the stand-alone radio equipment or the nominal voltages of the combined (host)equipment or test jig in case of plug-in devices:

Details provided are for the: ☒stand-alone equipment
☐combined (or host) equipment
☐test jig

Supply Voltage ☐AC mains State AC voltage _____ V
☒DC State DC voltage 3.3 V

In case of DC, indicate the type of power source
☐Internal Power Supply_____.
☐External Power Supply or AC/DC adapter:_____.
☐Battery:_____.
☒Other:DC 3.3V.

o) Describe the test modes available which can facilitate testing:
Continuous transmiting&receiving and normal operation.

p) The equipment type (e.g. Bluetooth®, IEEE 802.11™ [i.3], IEEE 802.15.4™ [i.4], proprietary, etc.):
Bluetooth®.

q) If applicable, the statistical analysis referred to in clause 5.4.1 q)

(to be provided as separate attachment)

r) If applicable, the statistical analysis referred to in clause 5.4.1 r)

(to be provided as separate attachment)

s) Geo-location capability supported by the equipment:

☐Yes
☐The geographical location determined by the equipment as defined in clause 4.3.1.13.2 or clause 4.3.2.12.2 is not accessible to the user.

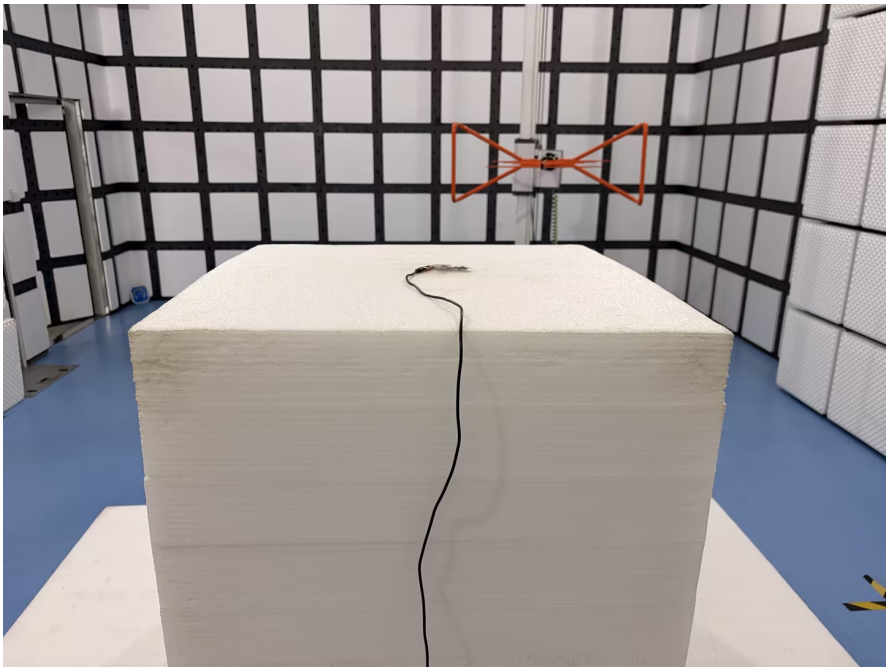
☒No

EXHIBIT B - EUT PHOTOGRAPHS

Refer to Report No.: 2607T13127E-EM-02

EXHIBIT C - TEST SETUP PHOTOGRAPHS

Radiated Spurious Emissions - Front View (Below 1GHz)



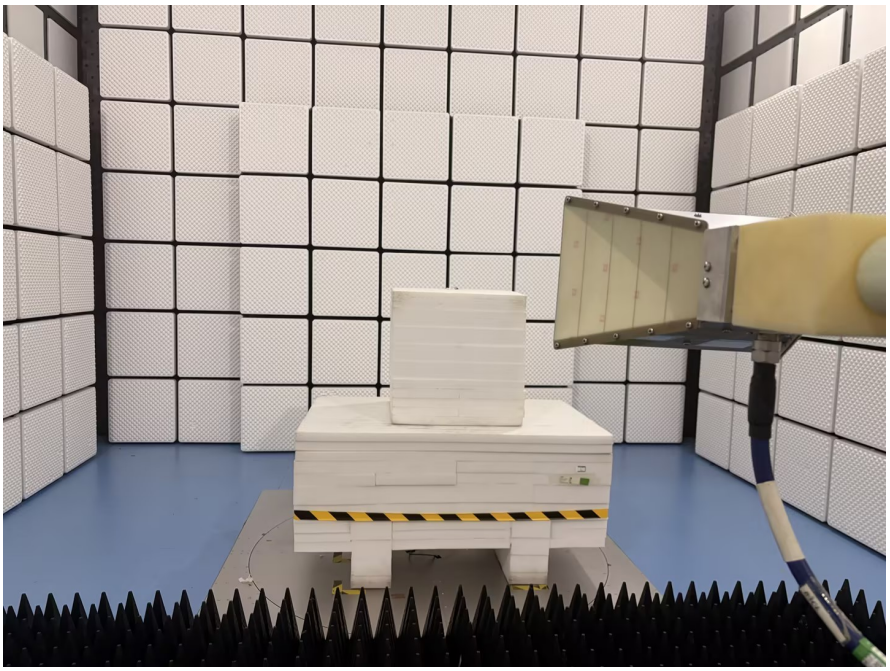
Radiated Spurious Emissions - Back View (Below 1GHz)



Radiated Spurious Emissions - Front View (Above 1GHz)



Radiated Spurious Emissions - Back View (Above 1GHz)



Declarations

1. Bay Area Compliance Laboratories Corp. (Fujian) is not responsible for authenticity of any information provided by the applicant. Information from the applicant that may affect test results are marked with an asterisk "★".

2. Unless otherwise stated the results shown in this test report refer only to the sample(s) tested.

3. Otherwise required by the applicant or Product Regulations, Decision Rule in this report did not consider the uncertainty.

4. The extended uncertainty given in this report is obtained by combining the standard uncertainty times the coverage factor K with the 95% confidence interval.

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******* END OF REPORT *******